



Introduction to Data Science-Week 1

Course Instructor : Mahrukh Khan

Introduction to the course

- Course Book
- Course Code(
- Course Outline
- Rules to clear this course
- Marks Distribution of Evaluations
 - Mid-Exams(25)
 - Final(40)
 - Project(10)
 - Assignments-3 (4 marks each)
 - Quizzes-4 (3 marks each)
 - Remaining - Attendance and Class Performance

Course Outline

- Introduction to Data Science, Data Science Lifecycle / /Types of data: Structured, semi-structured, unstructured, Data Types and sources (Kaggle, UCI, APIs)/Data acquisition methods (APIs, scraping basics)
- Exploratory Data Analysis with Python/Data Visualization with Python/Probability Concepts/Inferential Statistics
- Introduction to Machine Learning/**Model Evaluation Metrics/Supervised and Unsupervised Learning/ Feature Extraction & Dimensionality Reduction**
- **Associative Learning**/Market Basket Analysis/Apriori algorithm
- Data Science Ethics & Applied Data Science
- Advance Applications in Data Science/**Google PageRank & Intro to Graph Analytics**

Computer Science, Data Science

- Computer scientists by nature don't respect data, they have been taught that the algorithm was the real thing and data was just meant to be passed through it.
- CS care about answers but data science care about whether the results make sense.
- Data scientists are comfortable with the idea that data has errors, but Computer Scientists chant "Garbage in, garbage out" a way to say "That's not my job"
- Nothing is ever completely True or False, while in CS it is.

Cloud Computing and Data Science

- Delivery of on-demand computing resources over the internet on a pay-for-use basis rather than storing locally.
 - Networks, Servers, Applications , Storage, Services and Data centers
 - Example: Google Drive, Dropbox
- Access most current software versions and more cost effective.
- Cloud computing powers data science by providing scalable storage, processing, and tools to turn big data into valuable insights anywhere, anytime.

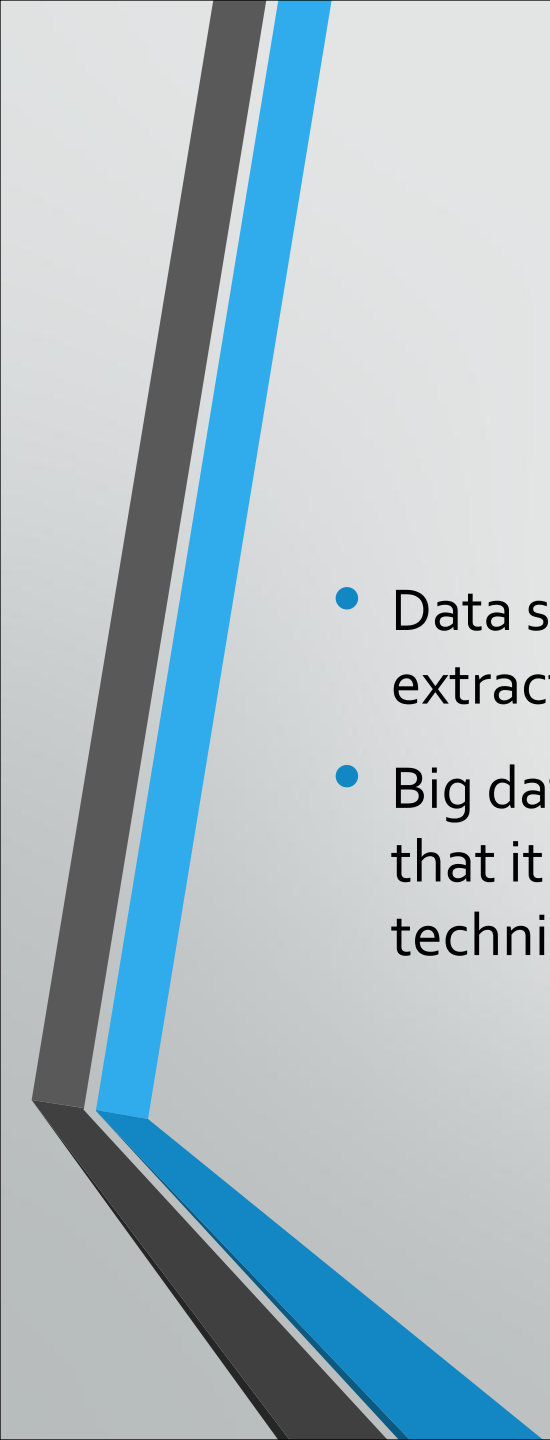


AI and Data Science

Data Science	It involves	It could use	It's a broad term
Is the process for extracting knowledge and insights from large volumes of data.	Mathematics Statistical analysis Data visualization Machine Learning And more	Machine Learning Algorithms Deep learning models	Encompasses the entire data processing methodology



AI includes everything that allows computers to learn how to solve problems and make intelligent decisions.



Data Science and Big Data

- Data science involves using methods to analyze massive amounts of data and extract the knowledge it contains.
- Big data is a blanket term for any collection of data sets so large or complex that it becomes difficult to process them using traditional data management techniques

Big Data

- In this digital world, everyone leaves a trace. From our travel habits to our workouts and entertainment, the increasing number of internet connected devices that we interact with on a daily basis record vast amounts of data about us.
- Big Data refers to the dynamic, large and disparate volumes of data being created by people, tools, and machines. It requires new, innovative, and scalable technology to collect, host, and analytically process the vast amount of data gathered in order to derive real-time business insights that relate to consumers, risk, profit, performance, productivity management, and enhanced shareholder value.

The V's of Big Data

Velocity

Volume

Variety

Veracity

Value



- The characteristics of big data are often referred to as the three Vs:
- Volume—How much data is there?
- Variety—How diverse are different types of data?
- Velocity—At what speed is new data generated?
- Veracity: How accurate is the data?

Big Data Processing Tools

- Big Data Processing Technologies provide ways to work with large sets of structured, semi-structured and unstructured data so that its value can be derived.(i.e. NoSql Databases and Data Lakes)



Big Data Processing

- The scale of the data being collected means that it's not feasible to use conventional data analysis tools. However, alternative tools that leverage distributed computing power can overcome this problem.
- Currently many big data tools and frameworks exist, and it's easy to get lost because new technologies appear rapidly. It's much easier once you realize that the big data ecosystem can be grouped into technologies that have similar goals and functionalities. Data scientists use many different technologies, but not all of them.



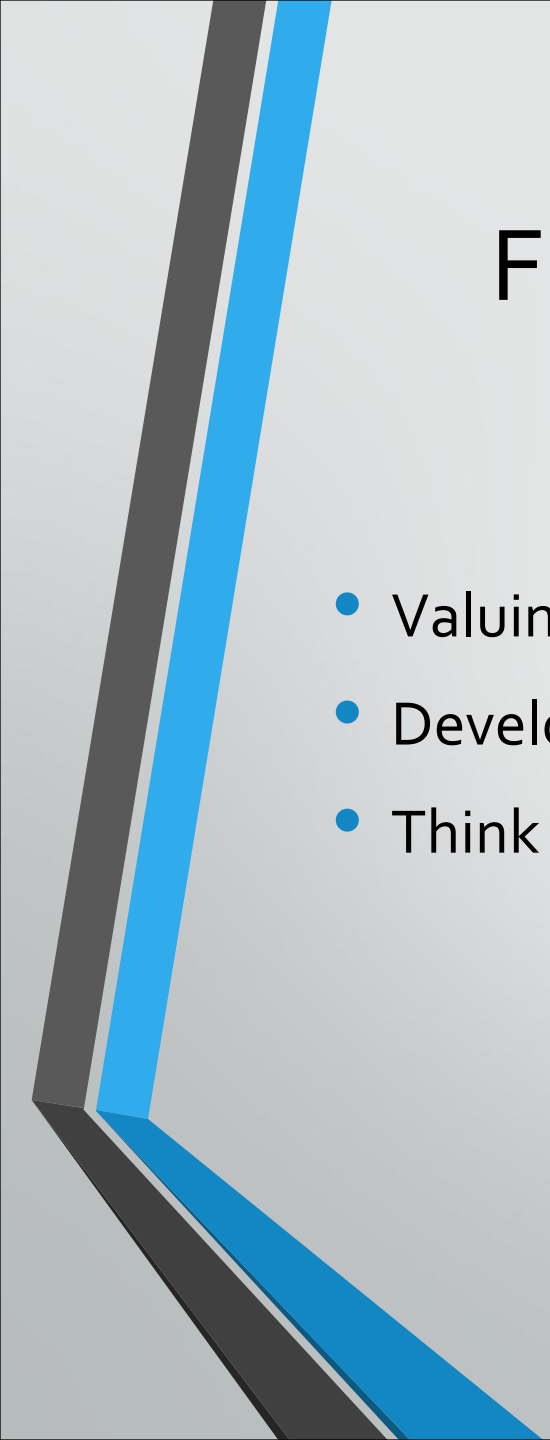
Apache Hadoop
a collection of tools that provides distributed storage and processing of big data



Apache Hive
a data warehouse for data query and analysis



Apache Spark
a distributed analytics framework for complex, real-time data analytics



Fundamentals of becoming a good data scientist

- Valuing doing the simple things right
- Developing mathematical intuition
- Think like a computer scientist, but act like a statistician

What will we do ???

- Given a problem , what available data will help us answer it???
- Given a dataset, what interesting problems can we apply it to ??

Applications of Data Science

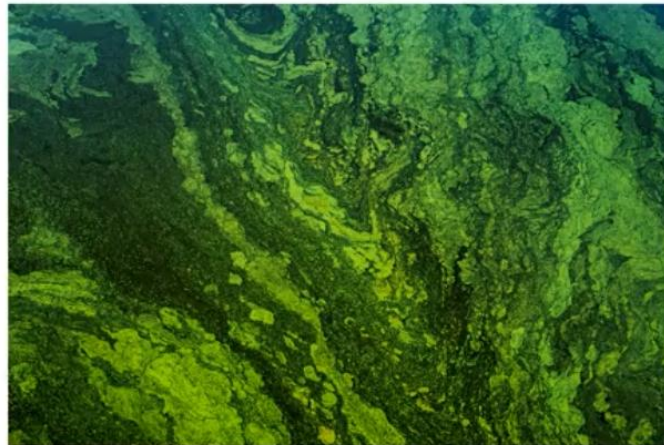
- Commercial companies in almost every industry use data science and big data to gain insights into their customers, processes, staff, completion, and products to offer customers a better user experience, as well as to cross-sell, up-sell, and personalize their offerings.
- The availability of vast amounts of data, and the competitive advantage that analyzing it brings, has triggered digital transformations throughout many industries. Netflix moved from being a postal lending system to one of the world's foremost video streaming providers. Organizations all around us are changing to their very core.
- A good example of this is Google AdSense, which collects data from internet users so relevant commercial messages can be matched to the person browsing the internet.

Example:

Data scientist may tackle environmental challenges such as predicting algae blooms to prevent water toxicity. By harnessing data in the prowess of artificial neural networks, data scientist can help water treatment companies safeguard the ecosystem.

Environmental problem investigation

Predicting algae blooms to prevent water toxicity



Helping water treatment companies safeguard the water

Asking interesting things from data-IMDb

- How do Hollywood movies compare to Bollywood movies, in terms of ratings, budget, and gross?
- What is the age distribution of actors and actresses in films? How much younger is the actress playing the mother, on average, than the actor playing the father? Has this disparity been increasing or decreasing with time?
- The technique of collaborative filtering finds people who liked films that I also liked, and give recommendations.
 - Netflix Prize to produce ratings engine better than the proprietary Netflix System.

Asking interesting things from data-Google Ngrams

- **Google Ngram Viewer** is a tool that lets you explore how often words or phrases appear in a huge collection of books over time.
- Google scanned **millions of books** (more than 8 million, published between the **1500s and 2019**) and turned the text into a searchable database.
- An “**ngram**” simply means a sequence of words:
 - **1-gram** = one word (e.g., “love”)
 - **2-gram** = two words together (e.g., “climate change”)
 - **3-gram** = three words (e.g., “machine learning model”)
- The Ngram Viewer lets you type in words/phrases and shows a **graph of their frequency over time**.

Activity

- How many fish are there in all the world's ocean?
 - How many people are flying in the air right now, all over the world?
 - How much tea is there in China?
 - How many gallons of milk are sold in Pakistan everyday?
-
- Make a quick guess based on your understanding, then use Google or Chatgpt to produce a more principled estimate form and see the difference.

Facets of Data

- Structured vs Unstructured
- Natural language vs Machine-generated
- Graph-based
- Audio, video, and images
- Streaming
- Quantitative vs Categorical

Data sources

- Video
- Audio
- Text

Data structure

- Structured
- Unstructured

Data formats

- Text files
- Spreadsheets
- XML
- PDFs
- JSON

Structured vs unstructured Data

- So structured data is more like tabular data things that you're familiar with in Microsoft Excel format. You've got rows and columns and that's called structured data.
- Unstructured data is basically data that is coming from mostly from web where it's not tabular. It is not, it's not in rows and columns. It's text. It's sometimes it's video and audio, so you would have to deploy more sophisticated algorithms to extract data.

1	Indicator ID	Dimension List	Timeframe	Numeric Value	Missing Value Flag	Confidence Inte
2	214390830	Total (Age-adjusted)	2008	74.6%		73.8%
3	214390833	Aged 18-44 years	2008	59.4%		58.0%
4	214390831	Aged 18-24 years	2008	37.4%		34.6%
5	214390832	Aged 25-44 years	2008	66.9%		65.5%
6	214390836	Aged 45-64 years	2008	88.6%		87.7%
7	214390834	Aged 45-54 years	2008	86.3%		85.1%
8	214390835	Aged 55-64 years	2008	91.5%		90.4%
9	214390840	Aged 65 years and over	2008	94.6%		93.8%
10	214390837	Aged 65-74 years	2008	93.6%		92.4%
11	214390838	Aged 75-84 years	2008	95.6%		94.4%
12	214390839	Aged 85 years and over	2008	96.0%		94.0%
13	214390841	Male (Age-adjusted)	2008	72.2%		71.1%
14	214390842	Female (Age-adjusted)	2008	76.8%		75.9%
15	214390843	White only (Age-adjusted)	2008	73.0%		73.0%
16	214390844	Black or African American only (Age-adjusted)	2008			
17	214390845	American Indian or Alaska Native only (Age-adjusted)	2008			
18	214390846	Asian only (Age-adjusted)	2008			
19	214390847	Native Hawaiian or Other Pacific Islander only (Age-adjusted)	2008			
20	214390848	2 or more races (Age-adjusted)	2008			

←
⏮
→
🗑 Delete
📁 Move
🛑 Spam
↑
↓
✕

● New team of UI engineers

● CDA@engineer.com

To xyz@program.com

Today 10:21 ★

An investment banking client of mine has had the go ahead to build a new team of UI engineers to work on various areas of a cutting-edge single-dealer trading platform.

They will be recruiting at all levels and paying between 40k & 85k (+ all the usual benefits of the banking world). I understand you may not be looking. I also understand you may be a contractor. Of the last 3 hires they brought into the team, two were contractors of 10 years who I honestly thought would never turn to what they considered "the dark side."

This is a genuine opportunity to work in an environment that's built up for best in industry and allows you to gain commercial experience with all the latest tools, tech, and processes.

There is more information below. I appreciate the spec is rather loose – They are not looking for specialists in Angular / Node / Backbone or any of the other buzz words in particular, rather an "engineer" who can wear many hats and is in touch with current tech & tinkers in their own time.

For more information and a confidential chat, please drop me a reply email. Appreciate you may not have an updated CV, but if you do that would be handy to have a look through if you don't mind sending.

↩ Reply
⏮ Reply to All
➡ Forward

Natural Language and Machine generated Data

- Natural language is a special type of unstructured data; it's challenging to process because it requires knowledge of specific data science techniques and linguistics involving entity recognition, topic recognition, summarization, text completion, and sentiment analysis

Machine-generated data is information that's automatically created by a machine without human intervention. Machine-generated data is becoming a major data resource and will continue to do so. (IoT)

The analysis of machine data relies on highly scalable tools, due to its high volume and speed. Examples of machine data are web server logs, call detail records, network event logs.

Quantitative vs Categorical

- Consists of numerical values like height and weight that can go directly into algebraic formulas and mathematical models
- Categorical data consists of label describing the properties like occupation hair color. It can be coded numerically

Graph data

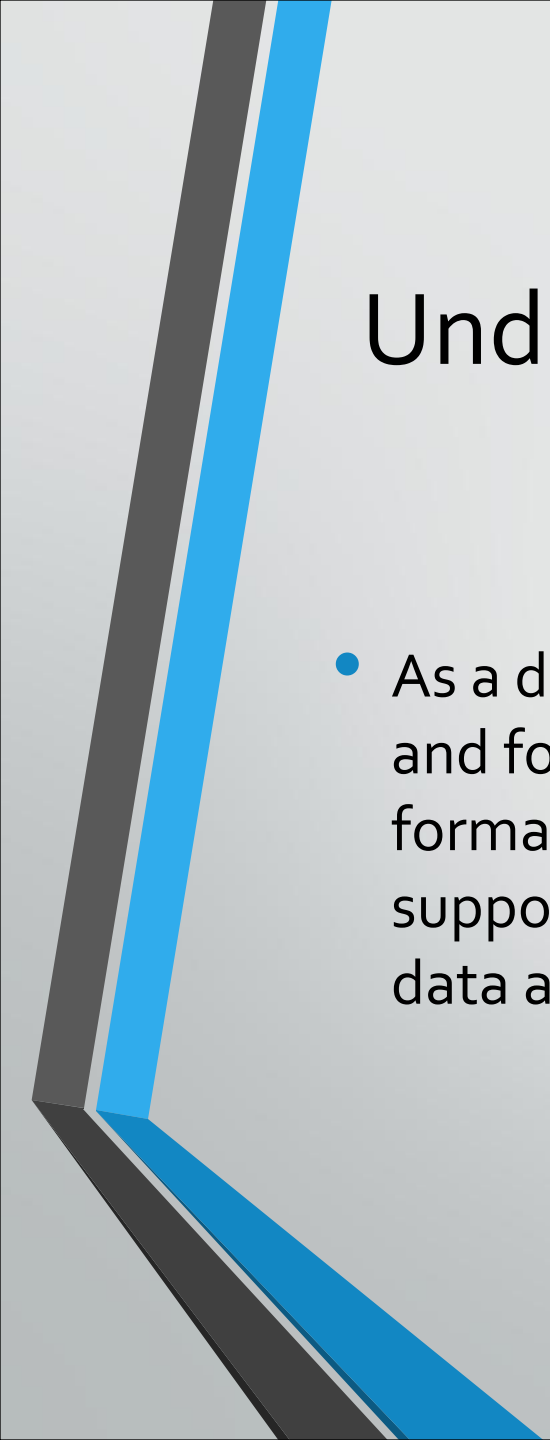
- In graph theory, a graph is a mathematical structure to model pair-wise relationships between objects. Graph or network data is, in short, data that focuses on the relationship or adjacency of objects.
- The graph structures use nodes, edges, and properties to represent and store graphical data. Graph-based data is a natural way to represent social networks, and its structure allows you to calculate specific metrics such as the influence of a person and the shortest path between two people.

Audio, Image and Video

- Audio, image, and video are data types that pose specific challenges to a data scientist.
- Tasks that are trivial for humans, such as recognizing objects in pictures, turn out to be challenging for computers.

Streaming Data

- While streaming data can take almost any of the previous forms, it has an extra property. The data flows into the system when an event happens instead of being loaded into a data store in a batch.
- Examples are the “What’s trending” on Twitter, live sporting or music events, and the stock market.



Understanding different types of File Formats

- As a data professional, you will be working with a variety of data file types, and formats. It is important to understand the underlying structure of file formats along with their benefits and limitations. This understanding will support you to make the right decisions on the formats best suited for your data and performance needs.

Standard File formats

1. Delimited text file formats, or .CSV
2. Microsoft Excel Open .XML Spreadsheet, or .XLSX
3. Extensible Markup Language, or .XML
4. Portable Document Format, or .PDF
5. JavaScript Object Notation, or .JSON

Delimited text files



Files used to store data as text

Each value is separated by a delimiter

Delimiter - A sequence of one or more characters for specifying the boundary between independent entities or values.

Comma, Tab, Colon, Vertical Bar, Space



Comma-separated values



Tab-separated values

Manufacturer, Model, Sales_in_thousands, __year_resale_value, Vehicle_type, Price_in_thousands

Acura, Integra, 16.919, 16.36, Passenger, 21.5

Acura, TL, 39.384, 19.875, Passenger, 28.4

Acura, CL, 14.114, 18.225, Passenger, 14

Acura, RL, 8.588, 29.725, Passenger, 42

Audi, A4, 20.397, 22.255, Passenger, 23.99

Audi, A6, 18.78, 23.555, Passenger, 33.95

Audi, A8, 1.38, 39, Passenger, 62

BMW, 323i, 19.747, Passenger, 26.99

BMW, 328i, 9.231, 28.675, Passenger, 33.4

BMW, 528i, 17.527, 36.125, Passenger, 38.9

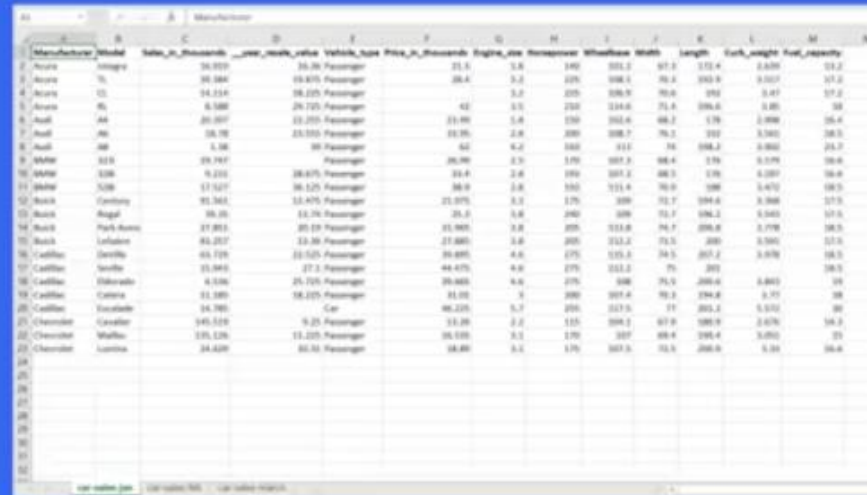
Buick, Century, 91.561, 12.475, Passenger, 21.975

.CSV

.TSV

Manufacturer	Model	Sales_in_thousands	__year_resale_value
Acura	Integra	16.919	16.36 Passenger 21.5
Acura	TL	39.384	19.875 Passenger 28.4
Acura	CL	14.114	18.225 Passenger
Acura	RL	8.588	29.725 Passenger 42
Audi	A4	20.397	22.255 Passenger 23.99
Audi	A6	18.78	23.555 Passenger 33.95
Audi	A8	1.38	39 Passenger 62
BMW	323i	19.747	Passenger 26.99
BMW	328i	9.231	28.675 Passenger 33.4
BMW	528i	17.527	36.125 Passenger 38.9
Buick	Century	91.561	12.475 Passenger 21.975

Microsoft Excel Open XML Spreadsheet, or .XLSX



The screenshot shows an Excel spreadsheet with the following data:

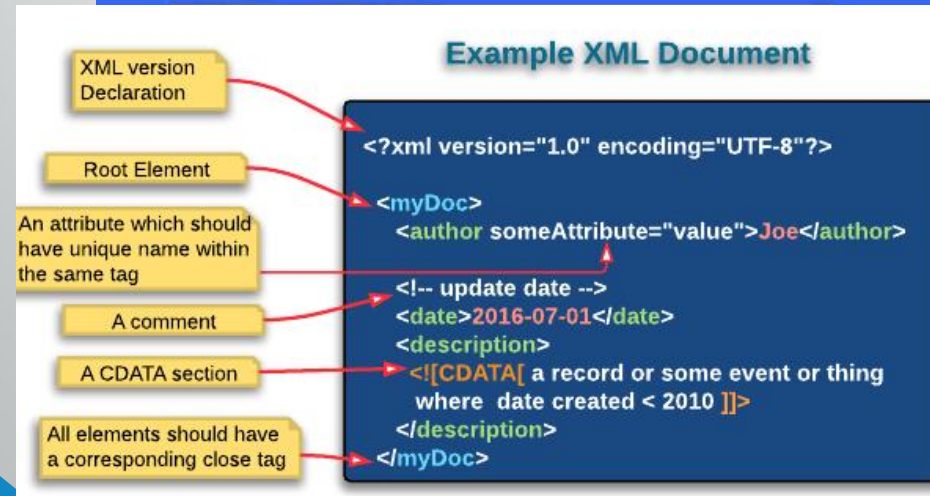
Manufacturer	Model	Sales_in_thousands	year	make	price	Vehicle_type	Price_in_thousands	Engine_size	Displacement	Wheelbase	Weight	length	Curb_weight	Fuel_capacity
Acura	Integra	36,929			25.5	Passenger	25.5	5.8	140	101.2	37.3	171.4	2,635	13.2
Acura	7c	30,384			25.4	Passenger	25.4	5.2	125	98.5	35.4	160.9	2,507	12.5
Acura	10	14,114			28.4	Passenger	28.4	5.2	125	106.5	36.6	162	2,477	12.2
Acura	4c	6,588			26.725	Passenger	42	5.0	120	114.6	31.8	156.6	2,485	38
Audi	A4	20,107			23.215	Passenger	23.85	5.8	130	102.6	48.2	178	2,899	26.4
Audi	A6	18,78			23.515	Passenger	23.35	2.8	100	108.7	36.2	172	3,340	28.5
Audi	A8	5,38			89	Passenger	62	4.2	142	111	56	198.2	3,850	25.7
Buick	1818	29,197				Passenger	26.36	2.5	170	107.3	48.4	176	3,178	36.6
Buick	126	9,232			28.475	Passenger	14.4	2.8	154	107.2	48.5	176	4,207	36.6
Buick	128	17,527			28.125	Passenger	38.9	2.8	150	111.4	36.9	188	3,470	38.5
Buick	Century	91,541			11.475	Passenger	21.075	3.2	175	109	12.7	194.6	3,368	17.5
Buick	Regal	39,35			11.24	Passenger	25.2	3.6	140	109	12.7	194.2	3,343	17.5
Buick	Park Avenue	27,851			25.19	Passenger	21.985	3.8	105	111.8	34.7	196.8	3,758	28.5
Buick	LeSabre	42,217			11.36	Passenger	27.485	3.8	105	111.2	35.5	200	3,565	17.5
Cadillac	Deville	65,128			11.525	Passenger	26.485	4.6	175	115.3	34.5	207.2	3,678	18.5
Cadillac	Seville	15,943			27.3	Passenger	44.475	4.9	175	112.2	35	203	3,675	18.5
Cadillac	Elkwood	4,536			25.715	Passenger	26.945	4.6	175	108	35.5	200.6	3,843	19
Cadillac	Catera	11,185			18.215	Passenger	31.02	5	100	107.4	35.2	194.8	3,77	38
Cadillac	Escalade	14,785				Car	46.225	5.7	105	117.5	71	201.2	5,152	30
Chrysler	Cavalier	345,118			9.25	Passenger	11.28	2.2	115	104.2	37.9	180.9	2,476	16.2
Chrysler	Stratus	135,126			11.225	Passenger	38.135	3.1	170	107	48.4	198.4	3,085	35
Chrysler	Luxon	14,428			30.31	Passenger	18.85	3.2	175	107.5	35.5	200.6	3,58	36.6

- Open file format, accessible to most other applications
- Can use and save all functions available in excel
- Is a secure file format as it cannot save malicious code

Extensible Markup Language or .XML

Extensible Markup Language, or XML, is a markup language with set rules for encoding data.

- Readable by both humans and machines
- Self-descriptive language
- Similar to .HTML in some respects
- Does not use predefined tags like .HTML does
- Platform independent
- Programming language independent
- Makes it simpler to share data between systems



Portable Document Format or PDF

The image shows a sample USAID form titled "APPLICATION FOR APPROVAL OF COMMODITY ELIGIBILITY (FORM AID-15)". The form is divided into several sections with numbered fields:

- Section 1:** USAID Letter of Commitment No. and Transaction No. (Assigned by USAID).
- Section 2:** Payment Terms, including U.S. Bank Letter of Credit No., Date, Name and Address of U.S. Bank (Advising Bank), and Other Payment Terms (if any).
- Section 3:** Export License No. and State.
- Section 4:** Supplier's Relationship to Authorized Source Country, with checkboxes for Corporation or Partnership, Individual, Other, and Unaffiliated Foreign Corporation.
- Section 5:** Supplier's Name and Address.
- Section 6:** Supplier's Name and Address.
- Section 7:** Contract, including Label Amount (Funded by USAID), Date, and Shipping Plans or Type of Application.
- Section 8:** Commodity Identification, including Commodity Description, Quantity, Size, and Unit and Unit Price, or Total F&B/Vol. Value Price, or F&B Price (Refined Port of Loading/Airport).
- Section 9:** Commodity Condition, with checkboxes for New and Unused, Used - Not Rebuilt or Reconditioned, Rebuilt, Reconditioned, and Other (Specify below).

Portable Document Format, or PDF, is a file format developed by Adobe to present documents independent of application software, hardware, and operating systems.

- Can be viewed the same way on any device
- Is frequently used in legal and financial documents
- Can also be used to fill in data for forms

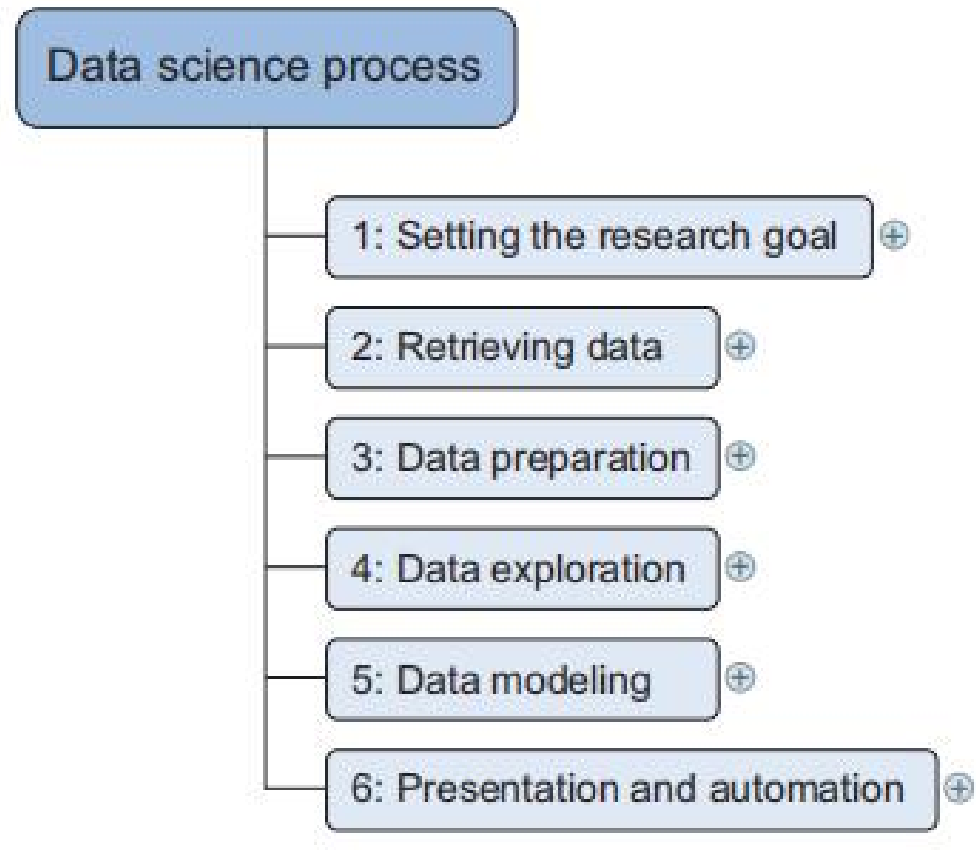
JavaScript Object Notation or JSON

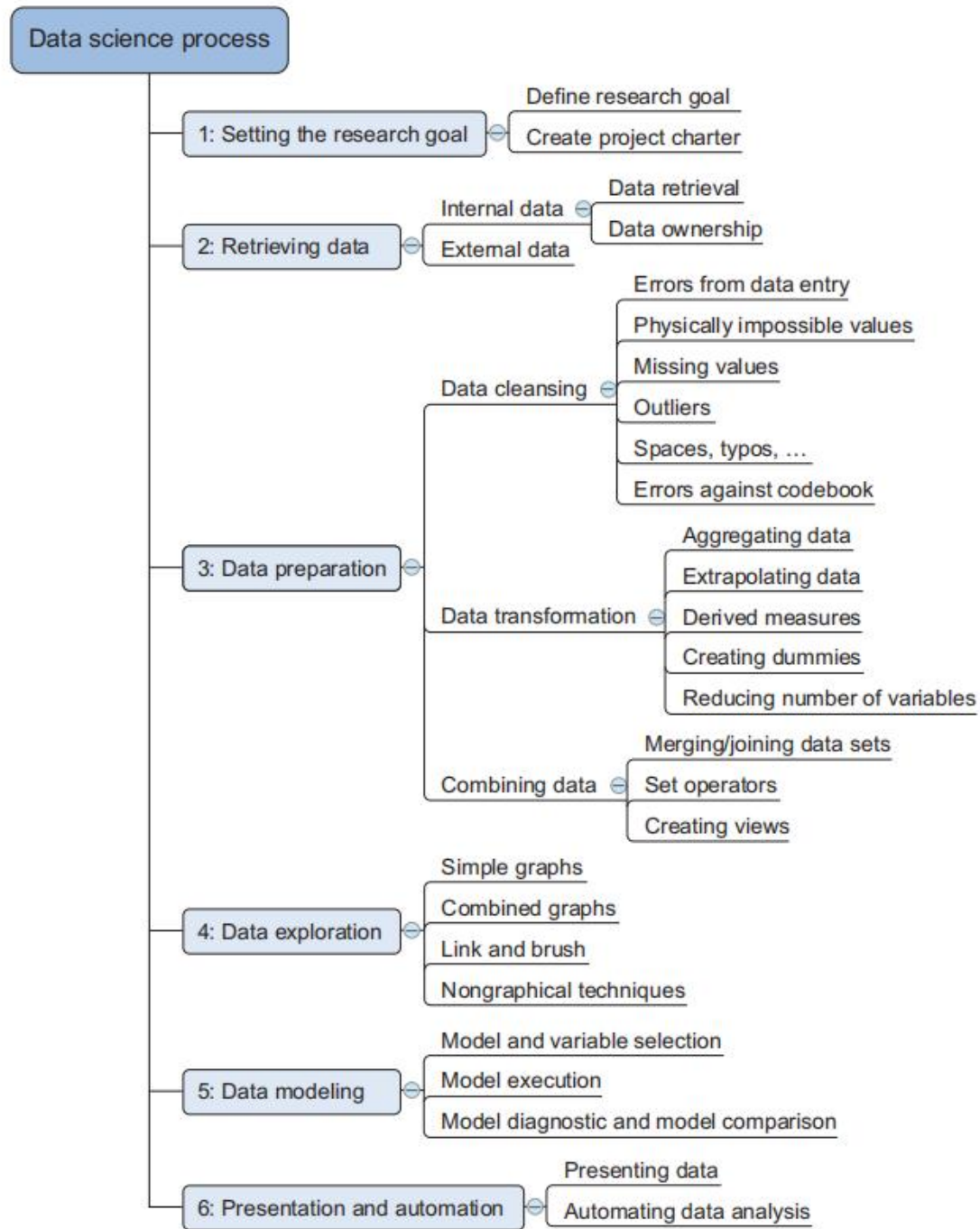
```
{  
  "Employee": [  
    {  
      "id": "1",  
      "Manufacturer": "Audi",  
      "Model": "Integra",  
    },  
    {  
      "id": "2",  
      "Manufacturer": "Buick",  
      "Model": "LeSabre",  
    },  
    {  
      "id": "3",  
      "Manufacturer": "Cadillac",  
      "Model": "Escalade",  
    }  
  ]  
}
```

JavaScript Object Notation, or JSON, is a text-based open standard designed for transmitting structured data over the web.

- Language-independent data format
- Can be read in any programming language
- Easy to use
- Compatible with a wide range of browsers
- Considered as one of the best tools for sharing data

the Data Science Process





Data Science and Statistics

- A data scientist uses statistics, but is not only a statistician. However, data scientists do use a lot of statistical models, such as statistical regression.
- Regression shows the probable relationship between two variables, such as the distance you drive and the amount of gas you use.

Regression-Example

Let me explain regression in the simplest possible terms. If you have ever taken a cab ride, a taxi ride, you understand regression. Here is how it works. The moment you sit in a cab ride, in a cab, you see that there's a fixed amount there. It says \$2.50. This is what you owe to the driver the moment you step into a cab. Then as it starts moving for every meter or hundred meters the fare increases by certain amount. And if you're not moving and you're stuck in traffic, then every additional minute you have to pay more. So as the minutes increase, your fare increases. As the distance increases, your fare increases. And while all this is happening you've already paid a base fare which is the constant. This is what regression is. Regression tells you what the base fare is and what is the relationship between time and the fare you have paid, and the distance you have traveled and the fare you've paid.